

Background

I recently started using clamAV on my main machine for malware scanning. But I had yet to do a full system scan, mainly using it to scan downloaded files during my job search. When I finally did run the full scan overnight, I woke up to a message showing 4 infected files were found. ClamAV doesn't log by default, so I didn't have a record of *which* files were considered infected. The next step? Run the scan again, send the output to a file that I can manipulate, and track down the files. From there we can investigate those files using some popular tools.

What is clamAV?

ClamAV is an antivirus software for the Linux operating system. There aren't many actively developed antivirus solutions for Linux that I could find. ClamAV is the most popular and operates from the terminal. This makes automating scans and refreshing the ClamAV rules very easy with things like **cron** or **systemd**.

Investigation

As mentioned before I ran another scan and sent the output to a file. While the scan was active I decided to use **grep** on the file to see if anything had been found yet. I first used **grep -v "OK"** to filter out all of the files that aren't counted as infected. After that I went through the output to see what I could find. There were a number of files classified as **Empty file** but there were also 4 entries marked as **FOUND**. Using **grep "FOUND"** will print only the infected files. The 4 files found were marked:

```
Win.Trojan.Gh0stRAT-9876493-0 FOUND
```

Three of these files were recovered from a hard drive many, many years before. The forth was found in a directory attached to an old game. To consolidate the information I directed the output to a file like so:

```
cat full.scan | grep "FOUND" > found.file
```

The output in **found.file** contains paths to the flagged files. After cleaning up the output I used the paths in **found.file** to move the files to a directory for quarantine:

```
cat found.file | xargs-I % mv '%' /.../quarantine
```

Once the files were moved I did a sha256 hash of each file and found the hash to be the same for all 4 files. From this I concluded that the recovered RAR files were copies of the game file. Searching the hash on VirusTotal confirmed this.

```
62f26b330a216cefded007d030d0f22a990e01d062e7f691648b5bf8bbe680e1 577641616-577643653_4.RAR
62f26b330a216cefded007d030d0f22a990e01d062e7f691648b5bf8bbe680e1 740096520-740098557_5.RAR
62f26b330a216cefded007d030d0f22a990e01d062e7f691648b5bf8bbe680e1 817012680-817014717_18.RAR
62f26b330a216cefded007d030d0f22a990e01d062e7f691648b5bf8bbe680e1 gamefile.rar
```

The screenshot shows the VirusTotal interface for the file 'gamefile.rar'. The top section displays a 'Community Score' of 4/61 and a warning that '4/61 security vendors flagged this file as malicious'. The file's SHA256 hash is 62f26b330a216cefded007d030d0f22a990e01d062e7f691648b5bf8bbe680e1, with a size of 1018.58 KB and a last analysis date of 7 years ago. The file is identified as a RAR archive. Below this, the 'DETECTION' tab is active, showing a table of security vendors' analysis results.

Security vendors' analysis		Do you want to automate checks?	
AegisLab	⚠ Packer.Genericic	Ikarus	⚠ Trojan.Retapu.D
Sophos ML	⚠ Heuristic	TrendMicro-HouseCall	⚠ PAK_Generic.005

Contents Metadata	
Contained Files	15
Uncompressed Size	4.20 MB
Earliest Content Modification	2012-02-08 13:07:38
Latest Content Modification	2013-03-30 17:59:09
Contained Files By Extension	
0	1
EXE	1
DAT	2
TXT	6

Only 2 vendors flagged the files as reported by VT. A search for **Trojan.Retapu.D** only returned results linking to pages that no longer existed. Using the Wayback Machine revealed the pages weren't archived either. However a search for the second result **PAK_Generic.005** did return a result from TrendMicro.

PAK_GENERIC.005

October 23, 2012

PLATFORM: Windows 2000, Windows XP, Windows Server 2003

OVERALL RISK RATING:

REPORTED INFECTION:

Low Medium High Critical

Threat Type: Trojan

Destructiveness: No

Encrypted:

In the wild: Yes

OVERVIEW

This is the Trend Micro detection for possibly malicious executable files that are compressed using Win32 compression tools. This proactive detection also includes appending viruses found in the wild.

From the investigation performed thus far, the file seems to be benign. The main concern seems to be that the files are RAR files and are commonly flagged. However, to be thorough I moved the game file into a VM to unpack the RAR and investigate further.

Unpacking

I moved the file from a shared folder directly into the VM's virtual disk and removed the other copies from my host machine. Using Microsoft Defender I scanned the decompressed files and nothing was flagged as malicious. At this point I could run each individual file through Virus Total and dig deeper, but after running the defender scan and further researching Gh0st RAT and I found no need to do so.

No current threats.

Last scan: 5/26/2025 5:43 PM (custom scan)

0 threat(s) found.

Scan lasted 1 seconds

55 files scanned.

Quick scan

Conclusion

ClamAV marks the files as a possible trojan named Gh0st RAT. This malware has many, *many* variants. Each variant is customized or written in a way that makes it hard to conclusively link them to each other. The number of variations and age of Gh0st RAT make false positives likely.

In a real enterprise/production environment the detection rules could be tweaked to avoid flagging *all* RAR files. Everything found during research suggests a very low risk level, and during triage would most likely be flagged as *low priority*. The findings also show no need for escalation and should be relatively quick to address in a real world scenario. My personal recommendation would be to ensure that games and other unauthorized software can't be downloaded onto company hardware without explicit approval. Security policies should be in place to address that concern and BYOD best practices should be implemented to mitigate the risk of legitimate malware making it's way into the organization's environment.

Further Reading

<https://www.trendmicro.com/vinfo/us/threat-encyclopedia/malware/GHOSTRAT>

<https://www.sentinelone.com/blog/the-curious-case-of-gh0st-malware/>